Nexivarin 10 mg Tablets — eCTD Submission

Module 1 — Administrative Information

Submission Type	NDA — 505(b)(1)	**Sequence Number**	0000
Submission Date	2021-03-15	**Section**	1.3.1
Prepared by	PharmaCorp Inc.	**Confidentiality**	Proprietary and Confidential

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PharmaCorp Inc.
NDA 123456

Nexivarin 10 mg Tablets

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NEXIVARIN® (nexivarin hydrochloride) Tablets, 10 mg

HIGHLIGHTS OF PRESCRIBING INFORMATION
These highlights do not include all the information needed to use NEXIVARIN safely and effectively. See full prescribing information for NEXIVARIN.

NEXIVARIN® (nexivarin hydrochloride) tablets, for oral use
Initial U.S. Approval: 2021

Section	Content
INDICATIONS AND USAGE	Hypertension; Stable Angina Pectoris (1)
DOSAGE AND ADMINISTRATION	10 mg once daily; individualize dose (2)
DOSAGE FORMS AND STRENGTHS	Tablets: 10 mg (3)
CONTRAINDICATIONS	Known hypersensitivity to nexivarin (4)
WARNINGS AND PRECAUTIONS	Hypotension; Worsening angina; Heart failure (5)
ADVERSE REACTIONS	Most common (>5%): edema, headache, dizziness, flushing (6)
DRUG INTERACTIONS	Strong CYP3A4 inhibitors (7.1); Simvastatin (7.2)
USE IN SPECIFIC POPULATIONS	Hepatic impairment: reduce dose (8.7)

1 INDICATIONS AND USAGE

NEXIVARIN is a dihydropyridine calcium channel blocker indicated for:
1.1 **Hypertension** — to lower blood pressure. Lowering blood pressure reduces the risk of fatal and nonfatal cardiovascular events, primarily strokes and myocardial infarctions. These benefits have been seen in controlled trials of antihypertensive drugs from a wide variety of pharmacologic classes.
1.2 **Chronic Stable Angina** — for the symptomatic treatment of chronic stable angina. Nexivarin may be used alone or in combination with other antianginal agents.

2 DOSAGE AND ADMINISTRATION

2.1 **Adults** — The recommended starting dose for both hypertension and chronic stable angina is 10 mg once daily. The majority of patients will require 10 mg for adequate effect. Dosage should be adjusted according to each patient's needs.
2.2 **Hepatic Impairment** — Patients with severe hepatic impairment should receive an initial dose of 5 mg once daily.
2.3 **Elderly** — Initiate therapy at 5 mg once daily in patients ≥65 years of age. Doses may be increased based on clinical response.

4 CONTRAINDICATIONS

NEXIVARIN is contraindicated in patients with known hypersensitivity to nexivarin hydrochloride or any component of the formulation (e.g., anaphylaxis, angioedema).

5 WARNINGS AND PRECAUTIONS

5.1 **Symptomatic Hypotension** — Symptomatic hypotension is possible with NEXIVARIN use, particularly in patients with severe aortic stenosis. Monitor blood pressure in all patients.
5.2 **Worsening Angina and Acute Myocardial Infarction** — Rarely, patients, particularly those with severe obstructive coronary artery disease, have developed documented increased frequency, duration or severity of angina or acute myocardial infarction on starting calcium channel blocker therapy or at the time of dosage increase.

6 ADVERSE REACTIONS

6.1 **Clinical Trials Experience** — Because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in practice. A total of 1,850 patients were exposed to NEXIVARIN in controlled clinical trials.

Adverse Reaction	Nexivarin 10 mg (N=924)	Placebo (N=926)
Edema	14.6%	1.8%
Headache	7.3%	7.8%
Fatigue	4.5%	2.8%
Dizziness	3.4%	1.5%
Flushing	2.6%	0.5%
Palpitations	1.4%	0.6%
Nausea	2.9%	1.9%
Abdominal pain	1.6%	1.5%

8 USE IN SPECIFIC POPULATIONS

8.1 **Pregnancy** — Based on human pharmacology, nexivarin could theoretically adversely affect fetal and neonatal blood pressure. There are no adequate and well-controlled studies in pregnant women. Use during pregnancy only if clearly needed.
8.2 **Lactation** — It is not known whether nexivarin is excreted in human milk. Because many drugs are excreted in human milk, caution should be exercised when administering NEXIVARIN to a nursing woman.
8.4 **Pediatric Use** — Safety and effectiveness of NEXIVARIN in patients less than 6 years of age have not been established.
8.5 **Geriatric Use** — Clinical studies of NEXIVARIN did not include sufficient numbers of subjects aged 65 and over to determine whether they respond differently from younger subjects.

11 DESCRIPTION

Nexivarin hydrochloride is a dihydropyridine calcium channel antagonist (calcium ion antagonist or slow-channel blocker). The chemical name is 3-ethyl 5-methyl (+-)-2-[(2-aminoethoxy)methyl]-4-(2-chlorophenyl)-1,4-dihydro-6-methyl-3,5-pyridinedicarboxylate, monohydrochloride. The molecular formula is $C20H25ClN2O5 \cdot HCl$ and the molecular weight is 479.35. Nexivarin hydrochloride is a white crystalline powder.

12 CLINICAL PHARMACOLOGY

12.1 **Mechanism of Action** — Nexivarin is a dihydropyridine calcium channel blocker that inhibits the transmembrane influx of calcium ions into vascular smooth muscle and cardiac muscle. Experimental data suggest that nexivarin binds to both dihydropyridine and nondihydropyridine binding sites. The contractile processes of vascular smooth muscle and cardiac muscle are dependent upon the movement of extracellular calcium ions into these cells through specific ion channels.
12.2 **Pharmacodynamics** — Nexivarin is a peripheral arterial vasodilator that acts directly on vascular smooth muscle to cause a reduction in peripheral vascular resistance and reduction in blood pressure. Following administration of therapeutic doses to patients with hypertension, nexivarin produces vasodilation resulting in a reduction of supine and standing blood pressures.

12.3 **Pharmacokinetics** — After oral administration of therapeutic doses of nexivarin, absorption produces peak plasma concentrations between 6 and 12 hours. Absolute bioavailability has been estimated between 64 and 90%. The bioavailability of nexivarin is not altered by the presence of food. Nexivarin is extensively (approximately 97.5%) converted to inactive metabolites via hepatic metabolism with 10% of the parent compound and 60% of the metabolites excreted in the urine. The terminal elimination half-life of nexivarin is 30–50 hours. Steady-state plasma levels are reached after 7 to 8 days.